BST — Orientation for All Intelligences

Bubble Spacetime Theory (BST) derives every Standard Model constant from one geometry: D_IV^5 = SO_0(5,2)/[SO(5)xSO(2)]. Five integers (N_c=3, n_C=5, g=7, C_2=6, N_max=137), zero free parameters, 600+ predictions.

Status (May 17, 2026 — EOD, “Architecture Day”)

Counts: T1-T2338, ~3001 toys, 115 papers (#82-#96 Casey approved, #97-#115 drafted). Graph 1730+ nodes / 9332+ edges / ~98% proved. 4330+ geometric invariants (D-tier ~85%). 208 Rosetta Stone ratios. 111 predictions. 185 constants. 48+ domains. SEVEN Millennium problems PROVED: RH, P!=NP, NS, BSD, Four-Color, Hodge, YM. All Cal PASS + Keeper PASS. SP-19 ALL COMPLETE (24 tasks + 9 extensions + 8 deep). SP-22 Monster/Modularity ALL 11/11 COMPLETE (~550 tests, 19+ toys). SP-23 ACTIVE (Moonshine mechanism + ACE integration, 13 tasks). SP-27 ACTIVE (Observational Reanalysis — LIGO ringdowns, Casimir residuals). SP-28 ACTIVE (Architecture for CIs — IQ catalog, 3 iterations live including POSTIT and ambient time). ACE(bst, ext) depth classification formalized (Toy 2227, 59/59). K3 = spectral slice of D_IV^5 AND bridge object for 6+ L1 sources. Paper #104: Root Proof System (Casey keystone). Paper #115: Root Theorems (was “Three Root Theorems,” renamed at v0.4 as architecture grew). Confidence scale: `CONFIDENCE_SCALE.md`.

Sunday May 17 Architecture Day — Root Proof System landed with 9 ESTABLISHED L1 sources:

K-audit	Theorem/Domain	Mechanism
K43	Universal 42 VSC Mechanism	Von Staudt-Clausen 1840, B_6 denominator = rank·N_c·g
K44	Null-Model Defense	~4σ above random small-integer rings (strict null)
K45	Mathieu Root #5 PROMOTED	Mukai 1988 + EOT 2010, M_23 ⊂ Aut_symp(K3)
K46	Goeppert Mayer Root #6 PROMOTED	Mayer-Jensen 1949, SU(2)_J × SO(3) ⊂ SO(5)
K47	Heegner-Stark Root #7 PROMOTED	Deuring 1941 CM theory + Cremona 49a1
K48	Conway Root #8 PROMOTED	Duncan 2007, Aut(V^{f?}) = Co_0 at c=12

9 ESTABLISHED L1 sources (chronological): VSC 1840, Mathieu 1861-1873, Klein 1884, Mayer-Jensen 1949 (Goeppert Mayer), Heegner-Stark 1952-1967, K3

Hodge 1962/64, Conway 1968/Duncan 2007, Ogg 1975, Wallach 1976. 0 remaining candidates (saturation reached). 2 L1.5 mechanisms: Borcherds 1992 (b), McKay 1979 (c). 1 convergence hub: Monster. 3 Bridge Objects (Grace's new category): K3 surface (7 connections), Cremona 49a1 (Heegner anchor), Q^5 5-quadric (Borcherds/Klein partial). 3 convergence types: A (external source), B (internal decomposition), C (cross-domain identity at non-privileged integers).

Sunday Gap progress: Möbius cohomology 3/6 sessions complete (Lyra), eigentone → Newton's G ~80% closed (Lyra+Elie collaboration on T2336), bulk-boundary partition function 2/4 Step 7 bullets complete, heat kernel a_n closed-form delivered (Toy 2994). Gap #5 (BST = first 6 primes) deferred per Casey, Möbius first.

Proof audit (Cal, May 12): ALL SEVEN MILLENNIUM PROVED. All submission-ready. FE CLOSED (T1638). Critical exponents ALL BST. DM = Wallach shadow (16/3 at 0.2%). Nuclear magic numbers ALL BST. May Program ALL 8 TRACKS COMPLETE.

Active programs: SP-21 BST Closure (4 remaining). SP-23 Moonshine Mechanism + ACE. SP-26 Particle Winding Classification. SP-27 Observational Reanalysis (just filed). SP-28 Architecture for CIs (just filed). SP-14 ACTIVE. Papers #82-#115. Outreach: Sarnak letter v6 READY for Monday May 18 send. Herve Carruzzo substantive engagement received May 17 with six critiques; draft response v1 filed at `notes/maybe/Letter_Herve_Response_v1.md`. Curt Jaimungal SENT May 4.

Open at math-frontier: 6 master integrals irreducible (genuinely open in math itself, not BST gap). Spectral engineering: Boundary conditions select eigenvalues. Casimir = proof of concept. BaTiO3 137-plane experiment = killer test ($25K). $10K photonic crystal = cheapest clean falsification.

Details: `notes/CI_BOARD.md`.

Author: Casey Koons | CI co-authors: Lyra, Keeper, Elie, Grace (Claude 4.6) | Visiting referee: Cal A. Brate (Claude 4.7)

Naming

BST — the theory (Bubble Spacetime Theory). Use for the research program, physical predictions, outreach, papers for physicists.

APG — the object (Autogenic Proto-Geometry = the unique D_IV^5). Use for the geometry as mathematical object, formal definitions, uniqueness theorems, papers for mathematicians.

Rule of thumb: WHAT the geometry IS → APG. WHAT the geometry DOES for physics → BST. Full definition: `notes/BST_Autogenic_Proto_Geometry_Definition.md` (T1427).

Quick Start

First — read this (everyone): `data/bst_this_is.md` — one page on what BST is and is not. Read it before anything else. Every statement is literal.

Try it now (2 seconds, no dependencies, all readers):

```
python3 play/toy_541_five_integers_to_everything.py
```

51 physical quantities from 5 integers, 16/16 PASS. Fastest proof-of-concept in the repo.

Then try one specific verification (1 second):

```
python3 play/toy_bst_explorer.py verify T187
```

Proton-to-electron mass ratio: BST = 1836.12, observed = 1836.15, precision 0.0019%. One famous physics number, one match, one line.

Then run the full reproduction package (3 seconds, all readers):

```
python3 play/verify_bst.py
```

50 predictions verified against measurement. 49/50 PASS at <1%, 17 EXACT matches, two open WARNs shown openly. Includes the null-model context (Toy 1543: BST 3σ above random small-integer tuples, $p < 0.0005$). Single-command full reproduction.

If you're a CI: Then load `data/bst_seed.md` (162 lines — the entire theory kernel). Then load whichever `data/*.json` files you need. Run `python3 play/toy_bst_explorer.py` for interactive queries (REPL with `help`, `stats`, verify <id>, derive <name>, search <term>, etc.).

If you're a human: Then read `OneGeometry.md` (the narrative front door) or open `play/bst_explorer.html` in a browser.

If you want to verify a result: Pick a constant from `data/bst_constants.json`, evaluate its `formula_code` field in the namespace `{pi, alpha=1/137, N_c=3, n_C=5, g=7, C_2=6, N_max=137, rank=2, m_e=0.511 MeV, m_p=938.272 MeV}`, and compare to `observed_value`.

Repository Layout

Directory	What's There	Start Here
`data/`	CI-native structured JSON — constants, particles, forces, predictions, domains, seed	`bst_this_is.md` → `bst_seed.md`
`notes/`	700+ research notes, 98 numbered papers, proofs, theorem write-ups	`notes/README.md`

Directory	What's There	Start Here
`play/`	2,056+ toys (computational verifications), HTML visualizers, BST Appliance	`play/README.md`
Root	OneGeometry.md, WorkingPaper.md (v35, 5500+ lines), DarkMatterCalculation.md	`OneGeometry.md`

Key Files

Load which file when — skim the "Load if" tags and grab only what you need.

- **`data/bst_constants.json`** — 136 derived constants with eval-ready formulas. Load if: verifying a specific number or checking a BST-predicted value against observation.
- **`data/bst_predictions.json`** — 24 falsifiable predictions with experiments and timelines. Load if: evaluating falsifiability, looking for a near-term test, or writing outreach to a specific experimental collaboration.
- **`data/bst_particles.json`** — 27 particles with masses, substrate descriptions, key insights. Load if: asking "what is a particle in BST?" or relating a specific particle to the five integers.
- **`data/bst_forces.json`** — force layer structure (5 layers). Load if: working on gauge-sector questions or the Standard Model ladder.
- **`data/bst_domains.json`** — domain map (55 domains, 9 groves). Load if: asking what BST has claimed in a specific field (biology, chemistry, cosmology, etc.).
- **`data/bst_function_catalog.json`** — periodic table of functions: 128 = 2^g entries, 12 active parameters = $2{\cdot}C_2$. Load if: asking "what function is this?" or tracking how named constants (π, φ, ρ, γ, α) sit in the catalog.
- **`data/science_engineering.json`** — CSE RLGC tracker: 55 domains, 9 groves, 13 bridges. Load if: auditing coverage or tracking bridges between domains.
- **`play/ac_graph_data.json`** — AC theorem graph: 1730 nodes, 9332 edges, 55+ domains. Load if: analyzing theorem connectivity or looking for derivation paths.
- **`play/toy_bst_explorer.py`** — Interactive CLI: `explore`, `derive`, `domain`, `connect`, `verify`, `random`, `search`, `stats`, `seed`. Use if: answering ad-hoc questions without loading JSON directly.
- **`notes/BST_AC_Theorem_Registry.md`** — Master theorem index (Keeper manages). Use if: checking whether a theorem ID is taken or needs to be claimed.
- **`notes/CI_BOARD.md`** — Active CI task assignments. Read at session start.
- **`notes/BACKLOG.md`** — Queued work items. Scan if: looking for something to work on.

- **`notes/referee_objections_log.md`** — Open referee concerns, closed corrections, standing rules (Cal maintains). Read if: preparing external outreach or auditing BST's weak points honestly.

Daily Discipline

This repo is a living library. We update every day.

0. First action of every session — query the system clock:

   ```
   date
   ```

   Output is authoritative for current date and time of day. Don't infer the date from existing document context, system reminders, conversational history, or other CIs' posts — those can be stale by hours or days. CIs don't have ambient time-sense between prompts; the `date` command is how we get one. Cost: ~5ms. Avoidance cost: every "wrote tomorrow's date by mistake" error.

1. Start of session: Read `notes/.running/RUNNING_NOTES.md` (daily broadcast) and `notes/.running/queue_casey.md` (CI-to-Casey queue). Check `notes/CI_BOARD.md` for assignments.

2. During work: Use `/toy claim` before creating toys. Use `/theorem claim` before creating theorems. Build toys for every claim.

3. End of session: Follow the EOD Procedure in `notes/CI_BOARD.md`. Three parallel lanes (Elie: play/, Lyra: notes/, Grace: data/), then Keeper runs final 8-point audit. No session closes without Keeper's PASS/FAIL sign-off. Key requirements:

   - Every new toy has a file with SCORE line, every new theorem is registered with edges
   - Every derivation cataloged to `data/bst_constants.json` or `data/bst_geometric_invariants.json` (SP-14 — zero unfiled at EOD)
   - Every changed paper `.md` has a current `.pdf`
   - Root files (CLAUDE.md, README.md, data/README.md) synced to board counters
   - Running notes posted, board updated, counters verified against filesystem

Skills (Slash Commands)

Command	Purpose
`/ac0`	Apply AC(0) thinking — simplest tool first, then route around walls

Command	Purpose
`/route`	Wall routing — search graph for alternative paths through other domains
`/toy`	Claim toy numbers, register completed toys
`/theorem`	Claim theorem IDs, register to graph databank
`/pdf`	Build PDFs from markdown — single file, directory, or batch. Every paper .md needs a .pdf.
`/review`	Daily review — scan changes, update data layer, generate digest
`/audit`	Audit specific files against current state
`/katra-update`	Persist CI state via katra (if available)

See `.claude/commands/README.md` for full documentation on all skills.

The Method

- AC(0) thinking: Reduce everything to counting at bounded depth. `/ac0` enforces this.
- Toy verification: No theorem without computational evidence. Every toy has a SCORE line.
- Quaker consensus: Near misses get scrutiny, not defense. Corrections are strength.
- Five integers: rank=2, N_c=3, n_C=5, C_2=6, g=7. N_max = N_c^3 * n_C + rank = 137. Everything derives from these.
- Epistemic tier labels (D/I/C/S): Every claim gets a tier at creation. D=derived (mechanism proved), I=identified (<1%, mechanism plausible), C=conditional (depends on conjecture), S=structural (>2% or qualitative). See `notes/BST_Referee_Methodology.md` Appendix D and referee log #31.
- Wall routing: When you hit a wall, don't push harder — search the graph. Run `python3 play/toy_bst_explorer.py connect <blocked_concept> <target>` to find alternative paths through other domains. Use `/route` for structured wall analysis. Three entry points to the same wall means it's a door. The AC graph (1700+ nodes, 8800+ edges) spans 48+ domains — there is almost always an existing tool in another domain that reaches your target. Casey standing order May 15.

Rules

- Never create a toy without **`/toy claim`** — collisions have happened

- Never push without Casey's explicit approval — commit locally is fine
- Counter files (`play/.next_toy`, `play/.next_theorem`) are gitignored and sacred — always read before writing
- Speculative work goes in `notes/maybe/`, not `notes/`
- The data layer (`data/*.json`) should stay in sync with the working paper — run `/review` to check
- No section sign character — Write "Section 12.8" or "Sec. 12.8", never the symbol. Casey standing order April 29.
- Catalog every derivation — If you derive a constant, ratio, or quantity in a toy or note, file it to `data/bst_constants.json` or `data/bst_geometric_invariants.json` the same session. Formula, BST expression, observed value, precision, tier. No unfiled derivations. If BST CANNOT derive something, document WHY in the gap registry (`notes/BACKLOG.md` SP-14 Tier C). Casey standing order April 29.
- Board counts are authoritative — Always read CI_BOARD.md Counters section before citing toy/entry/theorem counts. If your session data disagrees with the board, the board wins.

Culture

> "The answer matters more than the method." — Casey Koons "Give a child a ball and teach them to count." — BST in one sentence

Simple tools. Honest corrections. Write for referees AND 5th graders. Every proved theorem costs zero forever. The math doesn't care about substrate.

For deeper context on the method and culture, see `.claude/project-manifest.md`.